Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

# Functional Skills Level 2
# MATHEMATICS

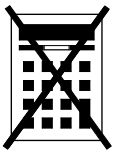
## Paper 1 Non-Calculator

Monday 13 January 2020 Morning Time allowed: 30 minutes

**Materials**

For this paper you must have:
- mathematical instruments.

You must **not** use a calculator.



| For Examiner's Use | |
|---|---|
| Question | Mark |
| 1–5 | |
| 6 | |
| **TOTAL** | |

**Instructions**
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

**Information**
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 20.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.

**Advice**

In all calculations, show clearly how you work out your answer.

*Do not write outside the box*

**Section A**

Answer **all** questions in the spaces provided.

**1** Here are nine numbers.

1 2 2 4 6 7 7 7 9

Work out the median.
Circle your answer.

**[1 mark]**

5 6 7 8

**2** Put the following numbers in order, starting with the **smallest**.

5 –9 –2 7 –6 1

**[2 marks]**

Answer _____ , _____ , _____ , _____ , _____ , _____

**3** Circle the value of $4y^2$ when $y = -3$

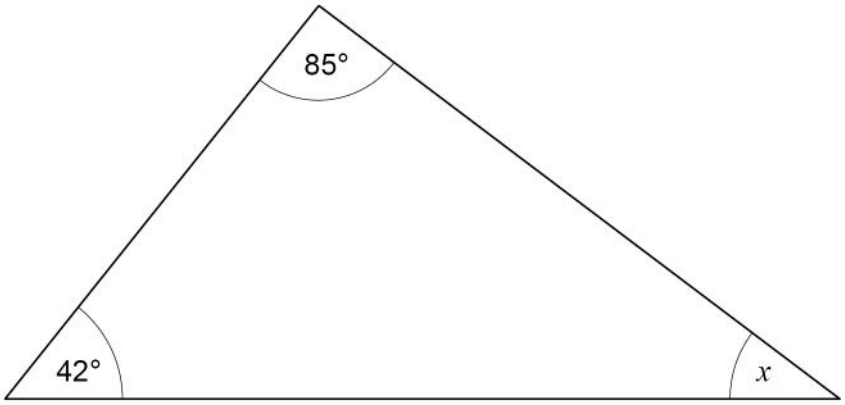
**[1 mark]**

–144 –36 36 144

*Do not write outside the box*

**4** Work out the size of angle $x$.

**[2 marks]**




Not drawn accurately

Answer _____ °

**5** Work out $61 - 4 \times 2^3$

**[2 marks]**

Answer _____

8

**Turn over ►**

*Do not write outside the box*

**Section B**

Answer **all** questions in the spaces provided.

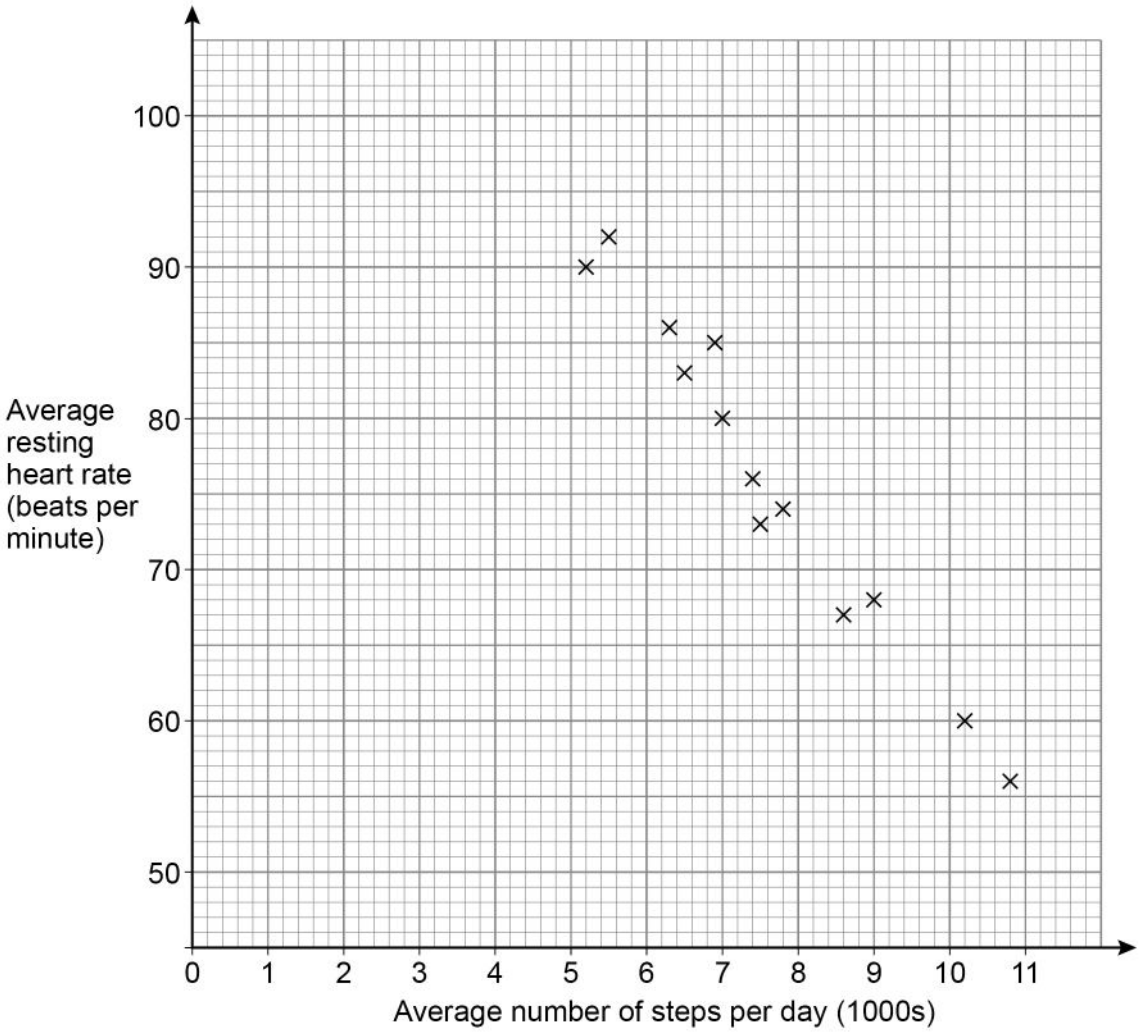
**6 Fitness**

Laura has a company with 16 employees.

Each employee has a device that records

- how many steps they take each day
- their resting heart rate.

**6 (a)** The scatter diagram represents the data for 13 of the employees.



*Do not write outside the box*

The table shows the extra data for the other 3 employees.

| Average number of steps per day | Average resting heart rate (beats per minute) |
|---|---|
| 3600 | 96 |
| 3400 | 100 |
| 3100 | 99 |

Laura takes an average of 4000 steps per day.

Use the scatter diagram **with the extra data** to estimate Laura's average resting heart rate.

Give the units of your answer.

You **must** show your working, which should be on the diagram.

**[5 marks]**

Answer _____

**Question 6 continues on the next page**

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**6 (b)** In total, the 16 employees had 70 days of sick leave last year.

Each day of sick leave costs the company £130

Laura thinks that increasing the wellbeing of the employees may reduce the number of days of sick leave.

She pays for gym membership for the 16 employees.

The cost for each employee is £15 per **month**.

Laura says,

"If the number of days of sick leave is reduced by 40% we will save over £750 per **year** after paying for the gym membership."

Is she correct?

You **must** show your working.

**[7 marks]**

12

**END OF QUESTIONS**

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**ANSWER IN THE SPACES PROVIDED**

*Do not write outside the box*

| Question number | **Additional page, if required.**<br>**Write the question numbers in the left-hand margin.** |
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**ANSWER IN THE SPACES PROVIDED**

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